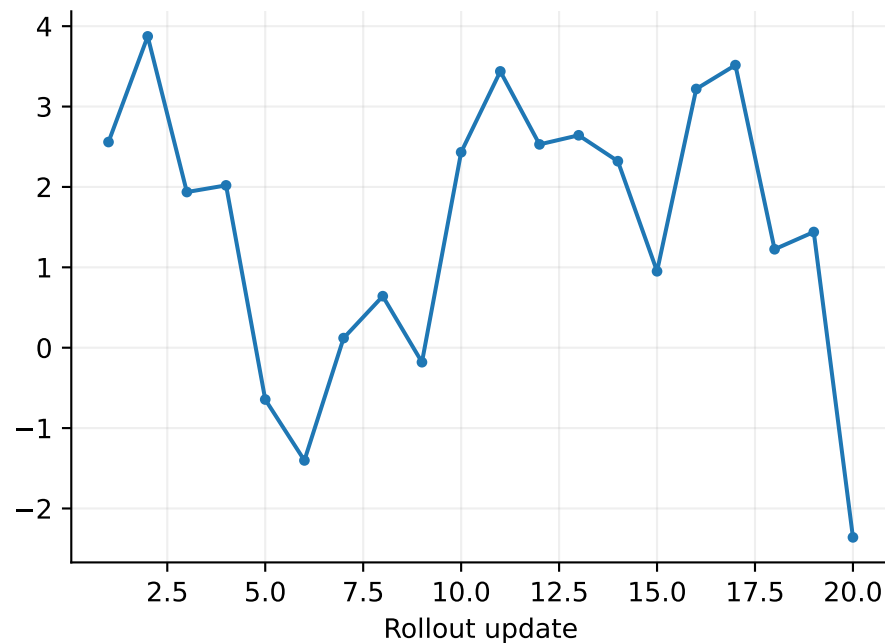
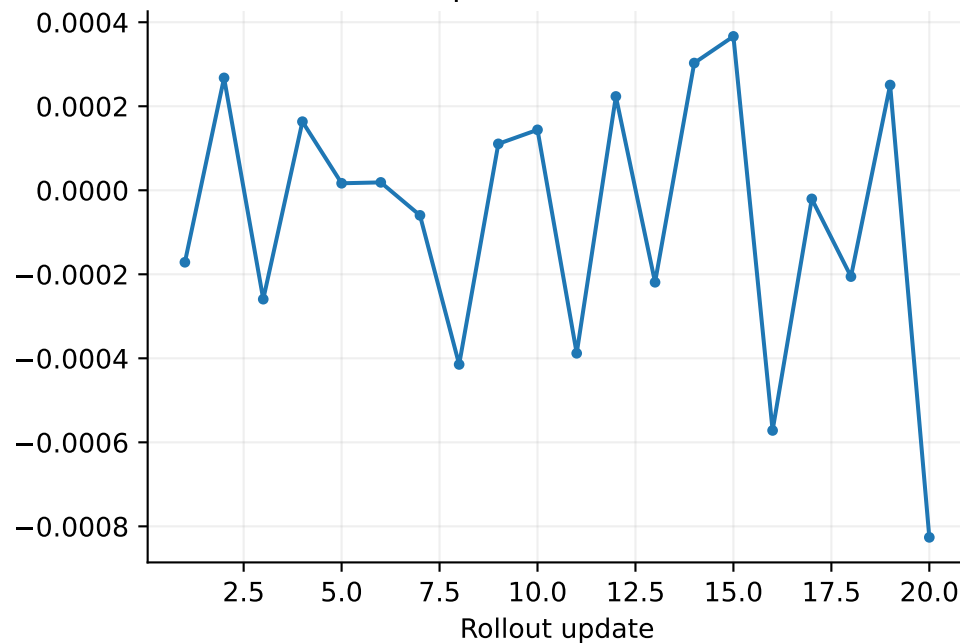


Standard PPO: 20 rollout updates, two optimization epochs each

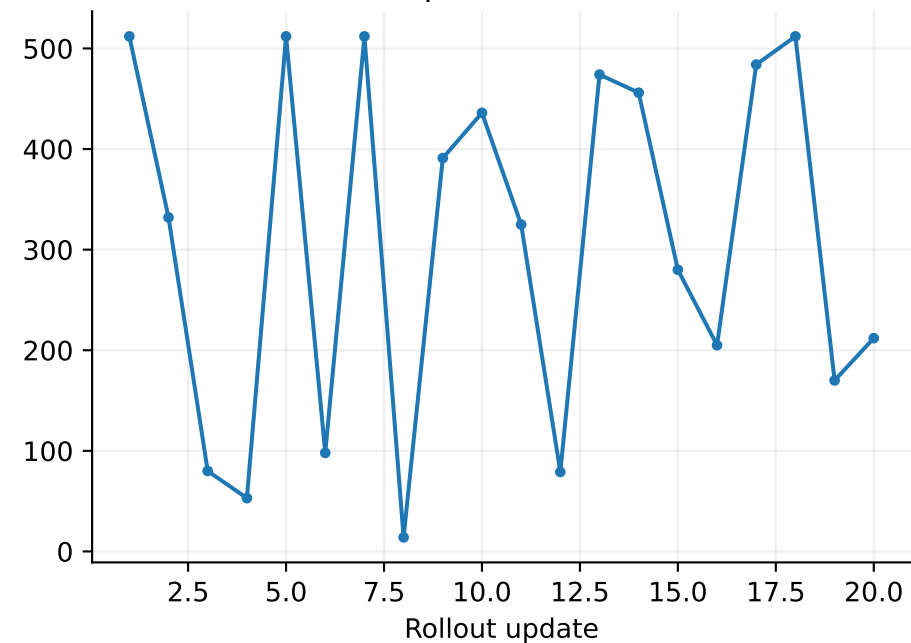
Raw learned reward



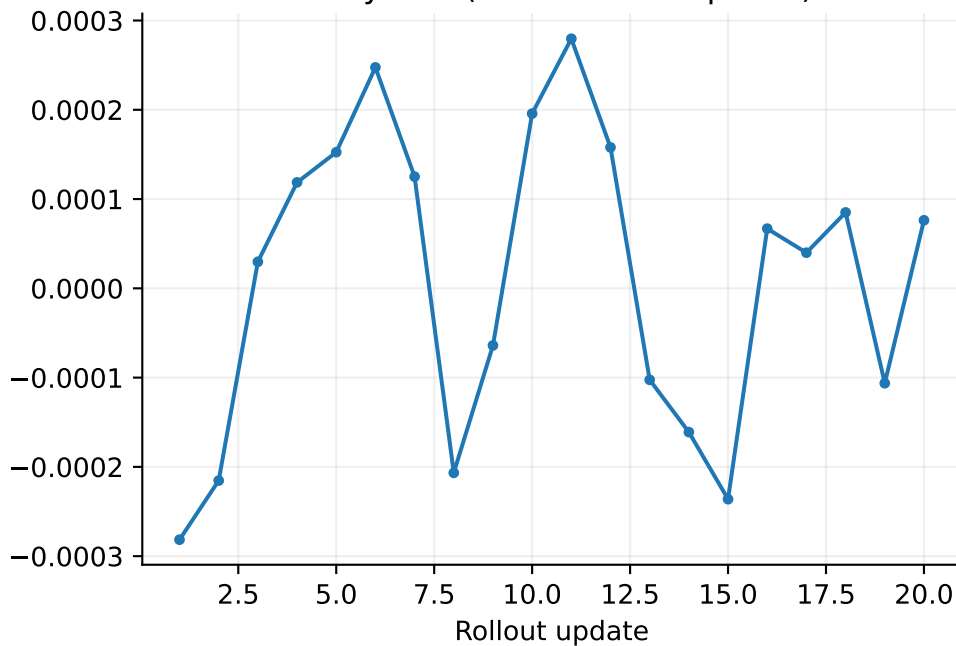
Sampled reference KL



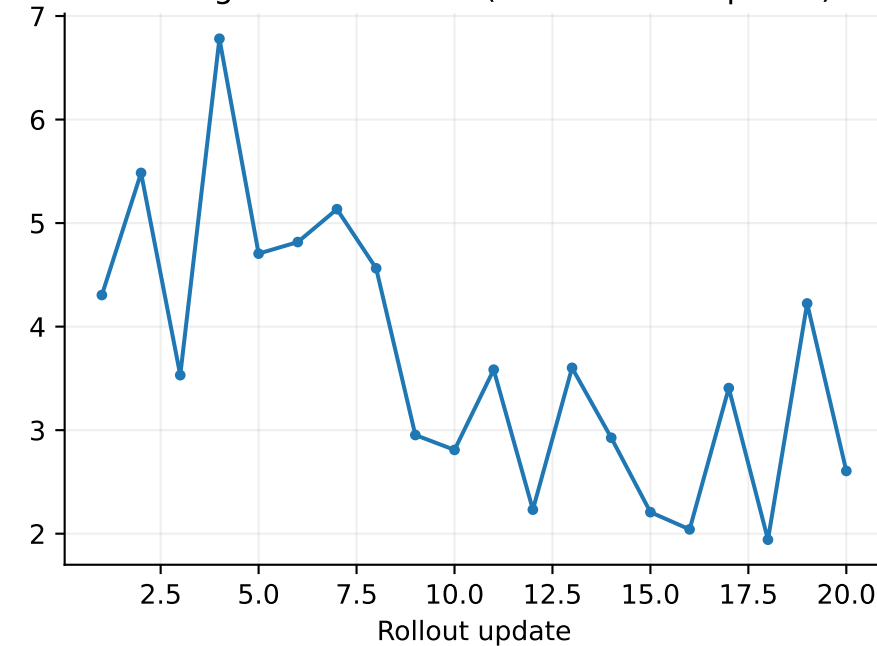
Response tokens



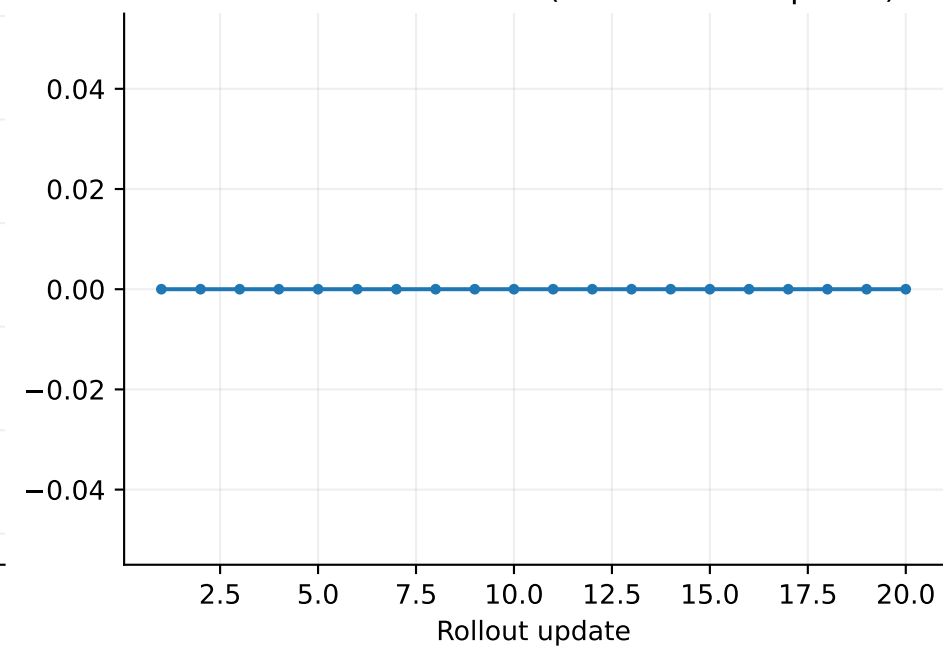
Policy loss (mean of two epochs)



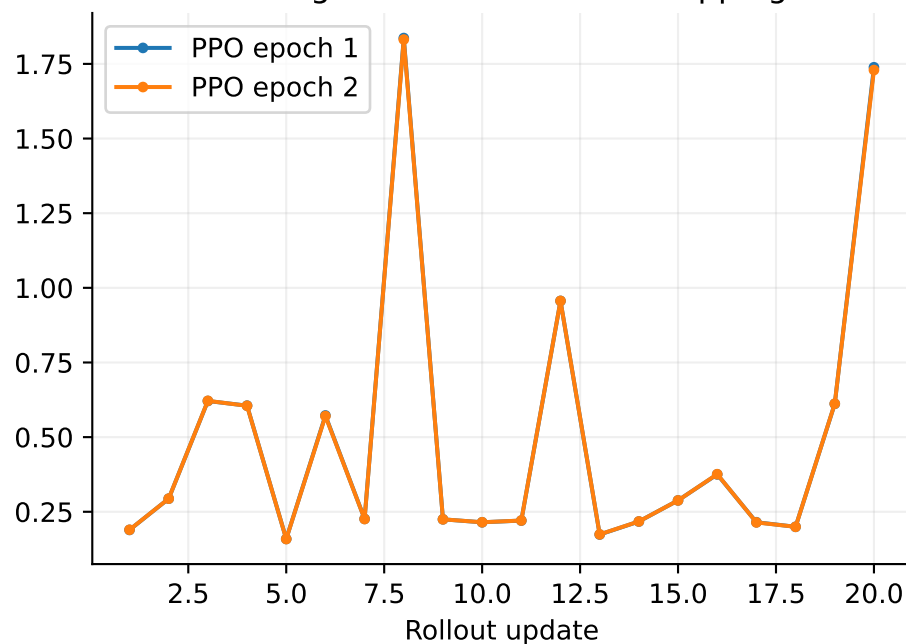
Unweighted critic MSE (mean of two epochs)



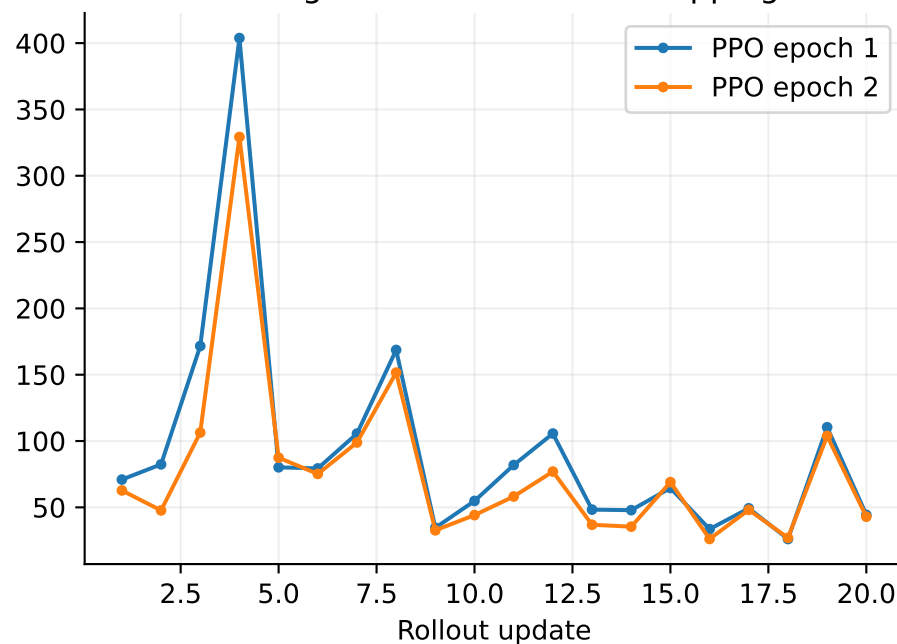
Affected-token fraction (mean of two epochs)



Actor gradient norm before clipping



Critic gradient norm before clipping



Rollout entropy (raw policy)

